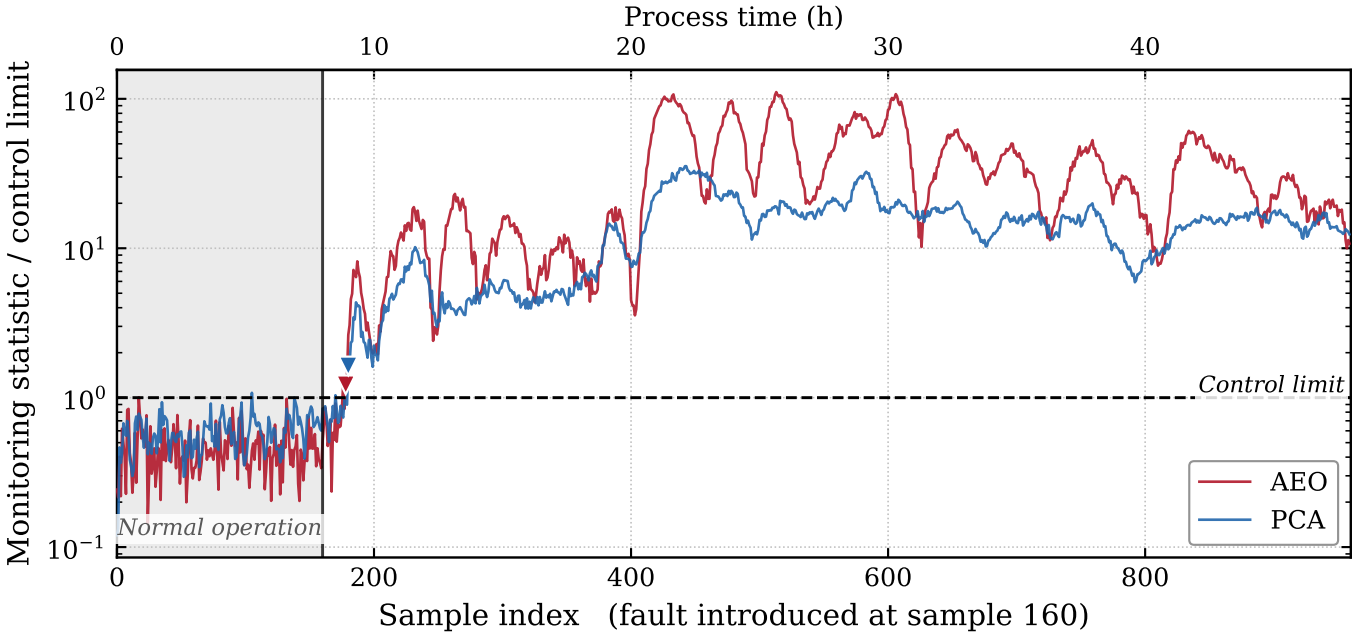
Fault Detection Performance for IDV(8)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	30.4 samples (91 min)	MDR 0.052	detected in 7/7 runs
PCA	delay	30.6 samples (92 min)	MDR 0.040	detected in 7/7 runs

Detection times comparable; PCA missed-detection rate lower by 24%